



Synthomer Chester SC (US)

Synthomer Chester Cooling Tower Performance Report - May 26

22 May 2026

Location

Synthomer Chester SC (US)

Prepared by

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System Health Report

Corrosion Control

The mild steel and copper corrosion rate are maintained well with average of 0.2 mpy respectively throughout the analysis period. Both values are well maintained within industry standards of <3.0 mpy for mild steel and <0.5 mpy for copper indicating effective corrosion inhibition and proper chemical control.

Scale Control

The polymer consumption of less than 25% confirms low scaling potential, suggesting minimal consumption of inhibitor actives and efficient scale prevention. The scale inhibitor concentration remained stable during the analysis period.

Microbial Control

The free residual chlorine is maintained well as 0.4 ppm which is within recommended standard. The baseline ORP remains relatively high at around 520 mV, and during dosing events, the ORP increases beyond 800 mV. This is resulting in huge spikes in copper corrosion. To ensure optimal biocide dosing based on actual system demand, the control strategy has been updated from timer-based dosing to timer with Delta ORP control. This approach allows the system to adjust dosing dynamically, ensuring that the required quantity of biocide is delivered based on real-time conditions rather than fixed durations.

Water Efficiency

The system is maintained with COC of 3.1 which is lower than the target cycles of concentration (COC) of 4.2. This decrease in conductivity indicates uncontrolled water losses; identifying and eliminating these will further stabilize cycle control and enhance overall water efficiency while minimizing unnecessary blowdown. This decrease in COC has resulted in water loss 2000 gpm per day. Now, the conductivity is well maintained with target COC of 4.2.

Product Efficiency

Product efficiency was well maintained with 100% efficiency throughout the period. The expected consumption was 96 lbs, which matched the actual consumption.

Proactive System Support:

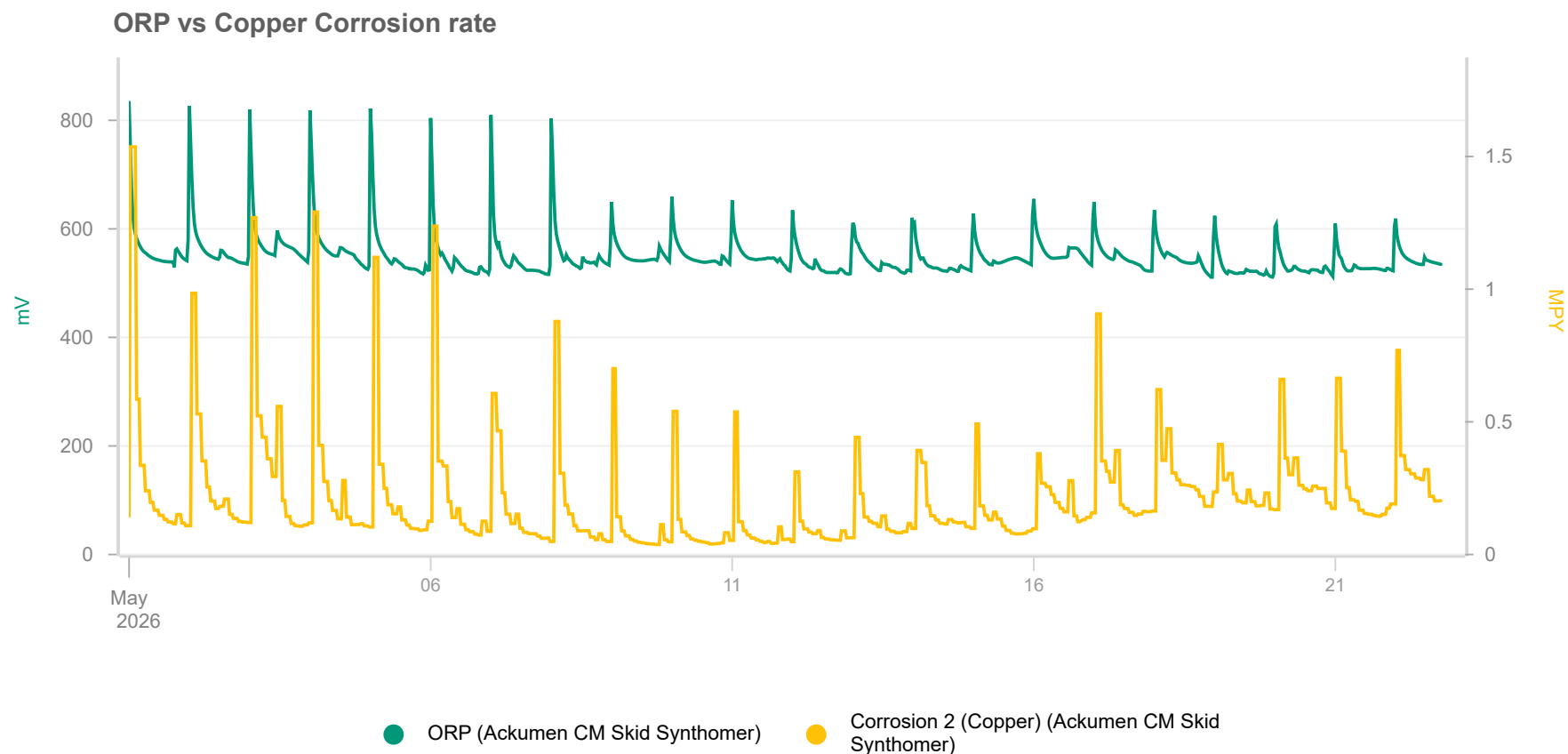
- This system has generated no alarms indicating the system is operating under good control.
- The biocide dosing control has been upgraded from timer-based operation to Delta ORP-based control to ensure dosing is aligned with real-time system demand. This improvement is expected to optimize biocide usage, reduce excessive ORP spikes, and minimize the impact on corrosion rates while maintaining effective microbial control.
- The waterloss has been addressed in service notes and the conductivity is well maintained at setpoint since service visit on 21/5/26.

Cooling Tower

Performance Summary

Parameter	Context Point	Average	Lower Limit	Upper Limit	% In Range	Gauge Chart
Corrosion 1 (Mild Steel)(MPY)	Ackumen CM Skid Synthomer	0.203	0.100	3.00	100	
Corrosion 2 (Copper)(MPY)	Ackumen CM Skid Synthomer	0.215	0.00	0.500	92.6	
Traced product(ppm)	Ackumen CM Skid Synthomer	106	50.0	200	100	
Tagged Polymer(ppm)	Ackumen CM Skid Synthomer	74.1	50.0	150	100	
pH(pH)	Ackumen CM Skid Synthomer	9.74	7.80	10.0	100	
ORP(mV)	Ackumen CM Skid Synthomer	550	200	600	93.6	
Electrode Conductivity(μS/cm)	Ackumen CM Skid Synthomer	889	600	1,500	100	
Cell Fouling(%)	Ackumen CM Skid Synthomer	7.42	-	30.0	100	

Cooling Tower Chart



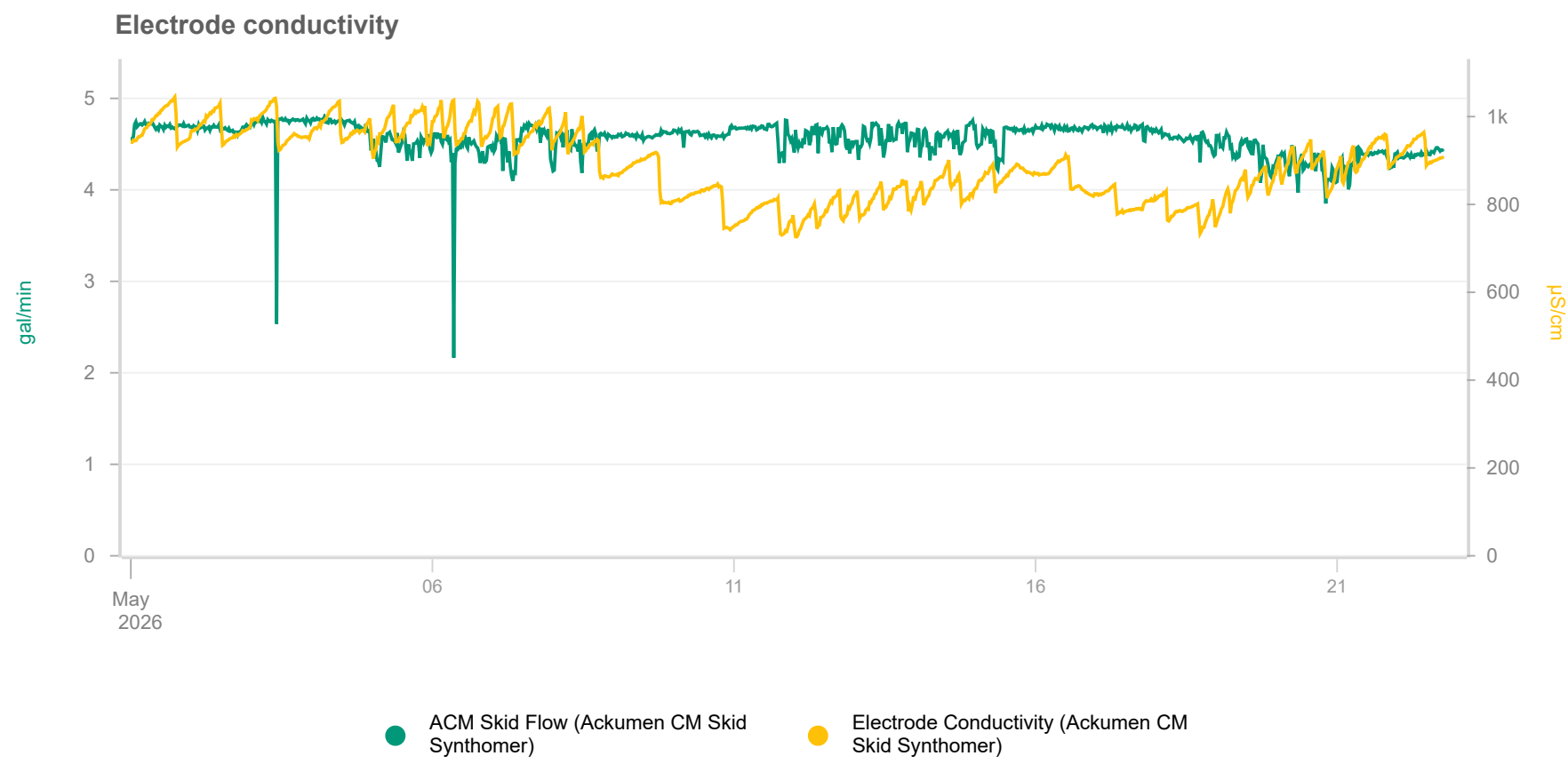
Comments

The baseline ORP remains relatively high at around 520 mV, and during dosing events, the ORP increases beyond 800 mV. The sharp increases in copper corrosion rates are observed during effective biocide dosing events, which can be attributed to the oxidizing nature of the biocide.

To ensure optimal biocide dosing based on actual system demand, the control strategy has been updated from timer-based dosing to timer with Delta ORP control. This approach allows the system to adjust dosing dynamically, ensuring that the required quantity of biocide is delivered based on real-time conditions rather than fixed durations.

After switching to Delta ORP-based control on May 8, biocide dosing became more stable and controlled, reducing ORP spikes. This led to lower copper corrosion rates while still maintaining effective microbial control.

Cooling Tower Chart



Comments

The system was initially maintained well at a COC of 4.2 and it started maintaining low as 3.1 since May 8. This reduction in conductivity reflected uncontrolled water losses, leading to approximately 2000 gpm lost per day. This has been addressed in service notes . Now, conductivity is consistently maintained at the target COC of 4.2.